

Latihan Soal UAP NLP

- **Main Menu**

The application menu **contains four menus**. **Validate** that the user can only select a number within the available **range of 1 to 4**.

```
Ril Or Fek
-----
Job Posting Classification
Real/Fake job posting classification & job recommendation based on text
Your text : None
Your text category : None
1. Write your texts
2. View Job Posting recommendation based on text
3. View Named Entities Recognition
4. Exit
```

Figure 1. Main Menu

- **Your text**

Text will be obtained from **user input**. When the text is empty, display ‘None’.

- **Your text category**

Text category will be obtained from **classifying the text from user input**. More details about the category will be explained in the **“Write Your Texts” menu**.

- If user choose **menu 1** (**“Write Your Texts”**), then:

- Users will be able to **input a text**. **Validate** that the input must **contain at least 20 characters and at least 3 words**.
 - Program will try to **load the model from file “naive_bayes_model.pkl”**. If the file **doesn’t exist**, train a new model using **Naïve Bayes** based on the dataset **“jobpostingdata.csv”** and save it with format name **“naive_bayes_model.pkl”**.

- When training the model, make sure to preprocess the dataset by **tokenizing the words**, **remove stop words**, **remove symbols and number**, and **lemmatizing the words**.

```
Input your text (minimum length 20 character):  
Model not found, training classifier ...  
Training classifier ...
```

Figure 2. Show that model is being trained when file is not found

- If the model is found, **classify the text** with **the model** and show the **classification result** of the text.

```
-----  
Job Posting Classification  
Real/Fake job posting classification & job recommendation based on text  
Your text : We are a big corporation that thrives for new technology and hiring for many new internships  
Your text category : Fake Job Posting  
1. Write your texts  
2. View Job Posting recommendation based on text  
3. View Named Entities Recognition  
4. Exit
```

Figure 3. Show the text and classification result of the text

- If user choose **menu 2** (“**View Job Posting Recommendation Based on Text**”), then:
 - **Check** whether there is a text or not
 - If there is no **text exist**, then show **message “No Text”** and redirect user **back to the main menu**.

```
No text  
Press enter to continue
```

Figure 4. Show 'No Text' when there is no text yet

- The application should support **one text representation** or **word embedding** techniques for **feature extraction**, you are allowed to choose from **TF-IDF**, **Word2Vec**, or **language models such as unigram, bigram, or trigram**. The application should use it to calculate **cosine similarities** between user text and text in the document.
- **Calculate the cosine similarity** between your text and the dataset.

- **Display the top 5 Job** from the **cosine similarity results** as job recommendations.

```

... Your Job Posting recommendation :

... 1    Financial Advisor Assistant
    2          Graduate Strategist
    3      Client Services Engineer
    4    Data Scientist (Part-Time)
    5    Senior Software Developer
    Name: title, dtype: object

... Enter to go back

```

Figure 5. Show top 5 most similar text title

- If user choose **menu 3** (“**View Named Entities Recognition**”), then:
 - The application will **validate** whether the user text exists or not. If user **text does not exist**, show ‘**No Text**’ and **redirect the user back to the main menu**.

```

Named Entity Recognition
No text

```

Figure 6. Show 'No Text' when there is no text yet

- The application will get the **entities** from user text using **spacy** library.
- **Show all named entities in the text.**

```

... Named Entity Recognition
    Named Entities
    PERSON: Barack Obama
    Press enter to continue

```

Figure 7. Show all named entity recognitions for every named entity in the text

- If user choose **menu 4** (“**Exit Menu**”), then terminate the program.